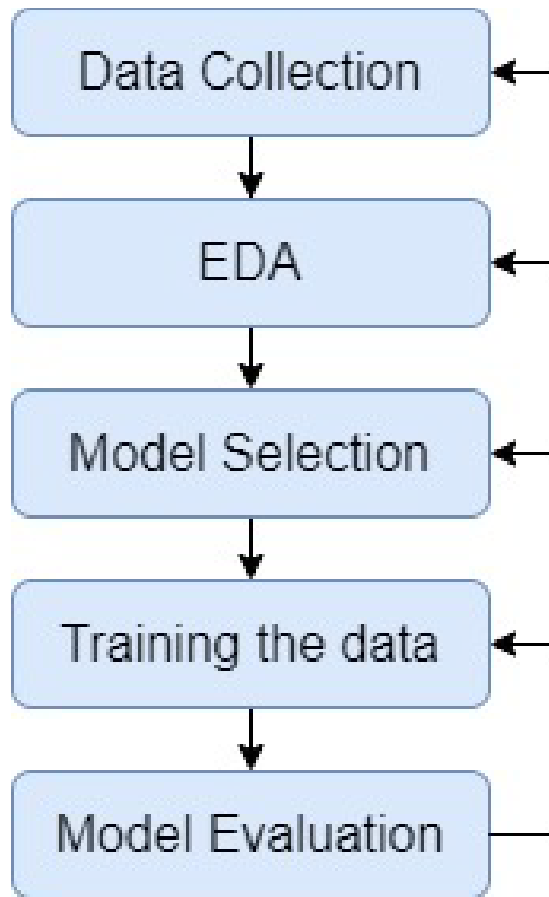


The problem of overfitting and underfitting and regularization

Learning Objectives

By end of this session, you should be able to:

Machine learning design cycle



Types of Error

There are 3 major errors that cause ML models to perform poorly:

- **Bias error:** Caused due to presumptions about the data.
- **Variance error:** Caused due to memorizing the data as opposed to learning.
- **Random error:** Caused due to the change unforeseen changes that we cannot do anything about.

Bias error

- Bias error means how far you have come in predicting the desired value from your actual value.
- It is an approach that can ultimately make or break the model in favor or against your idea.
- A straightforward example can be:

When we talk about the famous linear regression model, we quantify the relationship between X and Y variable as linear; on the contrary, in reality, the relationship might not be perfectly linear as we had read.

- It is always dependent on the quality and size of training data used to teach your model.

Variance error

- It is called the variance when your model performs exceptionally well on the training dataset yet fails to live up to the same standards when running it on an entirely new dataset.
- A model with low variance means that the sampled data is closer to the actual/expected data.
- Linear regression, logistic regression are few algorithms with low variance, whereas decision trees and SVM's are examples of algorithms with high variance.

Why data should not have bias?

- The first and most crucial factor is bias leading to misinterpretation of the data.
- It can stem from various sources, including humans who would have used uninformative data, incomplete surveys, and biased reports and measurements.
- The influence of biased data in Artificial Intelligence is well known.
- Example: There was a case in London where 'Alexa' started playing songs of her own will and saying discriminatory stuff, which further stresses that discrimination can take the shape of data.

Why data should not have variance?

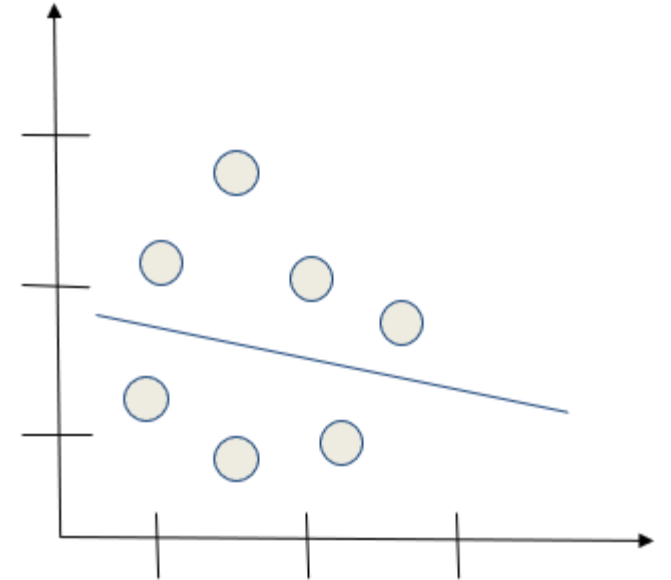
- One major disadvantage of variance is that it cannot be easily comprehended or interpreted.
- For example: In finance teams, variances can lead to tough questions on the credibility of the business and finance reports.
- Setting standard problems, specifications and approximations are fetched from different sources or even affected by human biases. This can lead to variance resulting in lousy information.

Bias vs Variance

- Bias leads to underfitting and variance leads to overfitting.
- Bias is the measure of difference between your predicted data from your actual data
- Variance is only the measure of how much your target data/function will change when a different training data is used.

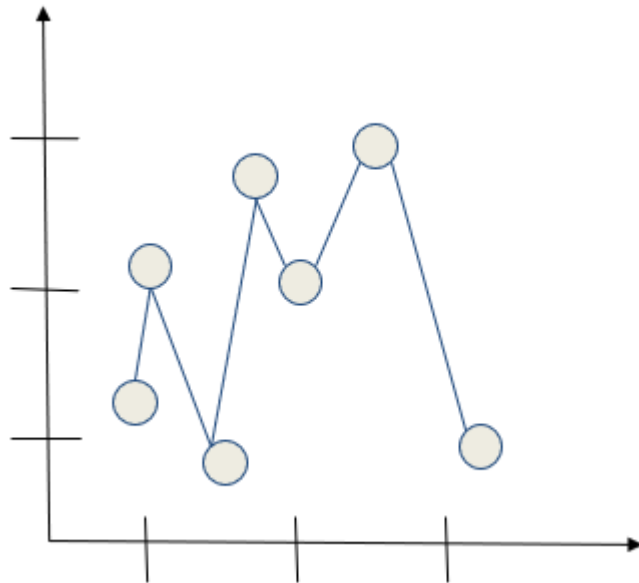
Underfitting

- When a statistical model or machine learning algorithm is unable to recognize the underlying trend in the data, it is said to have underfitting.
- It usually occurs:
 - While trying to avoid overfitting by stopping the training data feed to the model at an early stage;
 - Due to insufficient features considered while feeding training data;
 - Due to high bias error.
- Underfitting can be avoided by feeding more training data to the model and by increasing the number of features.



Overfitting

- When a statistical model or machine learning algorithm has been trained with a large amount of data, it is said to have overfitted.
- A model begins to understand the noise and erroneous data entries in our dataset when it is trained with such a large amount of data.



Terms to understand

Let's understand some important terms:

- **Low Bias:** This typically proposes fewer assumptions related to the target function.
- **High Bias:** This typically proposes more assumptions related to the target function. They are hinting towards an inaccurate prediction.
- **Low Variance:** This means that the sample data is closer to the predicted data, and there are slight changes in the target function of the training dataset.
- **High Variance:** This means that the sample data is further away from the predicted data, and there are considerable changes in the target function of the training dataset.

Under and Overfitting wrt Bias and Variance

Underfitting

High Bias

High Variance

Training Accuracy ↓

Testing Accuracy ↓

Overfitting

Low Bias

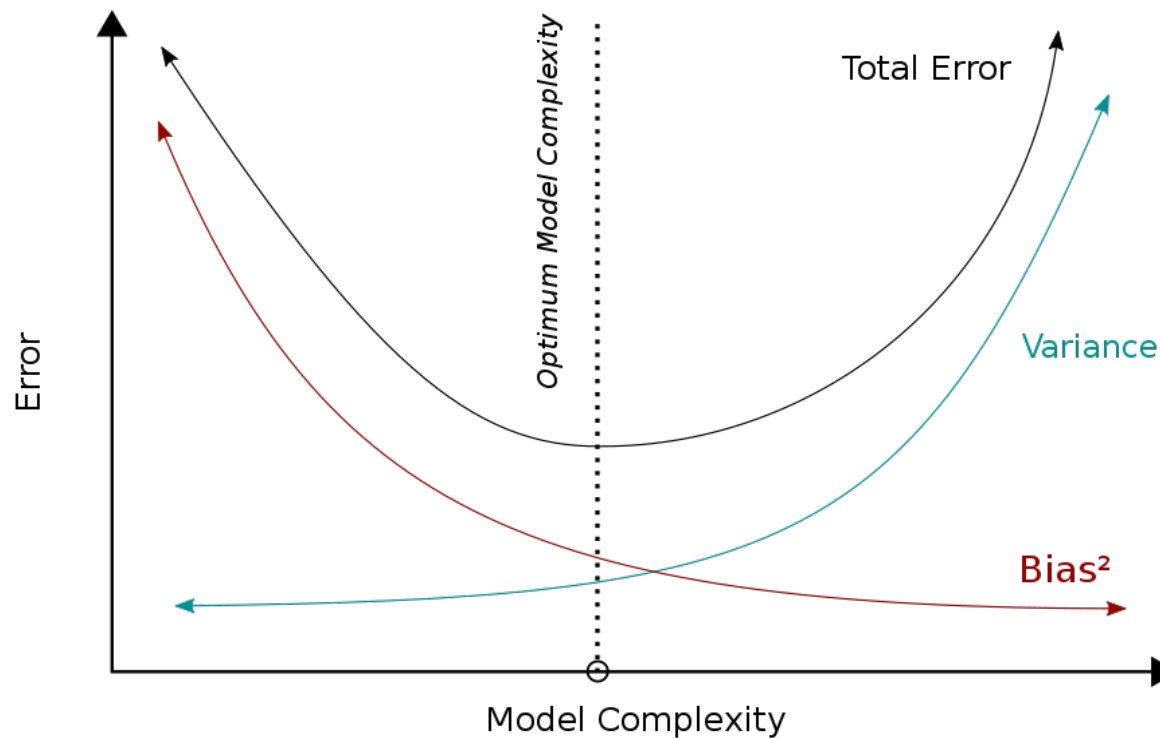
High Variance

Training Accuracy ↑

Testing Accuracy ↓

Bias Variance Trade Off

- This tension between bias and variance where the value of both bias and variance are the lowest is called as bias variance trade off.



Techniques to overcome overfitting and underfitting

Overfitting

- Increase the training data.
- Reduce model complexity
- Introduce early stopping during training phase
- Ridge or Lasso regularization
- Use dropout technique for neural networks.

Underfitting

- Make the model more complex
- Increase the number of features and performing feature engineering.
- Remove noise from the data.
- Increase the number of epochs or increase the duration of training to get better results.

Regularization

- Regularization is a regression type in machine learning background that is used for regularizing or shrinking the coefficient approximation to zero.
- In layman's terms this technique/methodology compels us not to learn a very complex or flexible model so that overfitting problem can be completely ignored.

When should you use regularization ?

- When there is high collinearity in machine learning model.
- When there is a low ratio of number of observations to number of variables.
- When the variables in consideration have high-dimensionality in the dataset.

Types Of Regularization

- There are two methodologies/techniques of regularization:
 1. Ridge Regression (L2)
 2. Lasso Regression (L1)
- The subtle difference between the two lies in the way in which they both assign penalty to the coefficients of the model.

Normalization

- When preparing data for machine learning, the scaling technique known as normalization is used to convert the values of the dataset's numerical columns to a standard scale.
- Normalization aims to scale down features to a similar scale.
- Although normalizing your data makes the training problem simpler, it does not directly prevent overfitting. For input data, such as photographs, when all input attributes have identical meanings, data normalization presents less of a challenge.

Learning Outcomes

You should now be able to: